Mathematical Specification: Mean-Variance Optimization with Ledoit-Wolf Shrinkage

Sacha Reymond

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1 The Primal Optimization Problem

We consider a universe of N assets. Let $\mathbf{w} \in \mathbb{R}^N$ be the weight vector, $\boldsymbol{\mu} \in \mathbb{R}^N$ the expected returns, and $\Sigma \in \mathbb{S}^N_{++}$ the covariance matrix. The long-only Markowitz problem is defined as:

$$\begin{aligned} \min_{\mathbf{w}} \quad & \frac{1}{2}\mathbf{w}^T \Sigma \mathbf{w} \\ \text{s.t.} \quad & \mathbf{w}^T \mathbf{1} = 1 \\ & \mathbf{w}^T \boldsymbol{\mu} = r_{target} \\ & w_i \geq 0, \quad \forall i \in \{1, \ldots, N\} \end{aligned} \tag{1}$$

2 The Lagrangian and KKT Conditions

We first derive the solution for the unconstrained case (without non-negativity constraints). The Lagrangian is:

$$\mathcal{L}_0(\mathbf{w}, \lambda, \gamma) = \frac{1}{2}\mathbf{w}^T \Sigma \mathbf{w} - \lambda(\mathbf{w}^T \mathbf{1} - 1) - \gamma(\mathbf{w}^T \boldsymbol{\mu} - r_{target}) \tag{2}$$

Taking the gradient with respect to $\mathbf{w}$ and setting it to zero (stationarity condition), we obtain:

$$\nabla_{\mathbf{w}} \mathcal{L}_0 = \Sigma \mathbf{w} - \lambda \mathbf{1} - \gamma \boldsymbol{\mu} = 0 \tag{3}$$

which implies:

$$\mathbf{w} = \Sigma^{-1}(\lambda \mathbf{1} + \gamma \boldsymbol{\mu}) \tag{4}$$

Substituting into the primal feasibility constraints $\mathbf{w}^T \mathbf{1} = 1$ and $\mathbf{w}^T \boldsymbol{\mu} = r_{target}$, we obtain:

$$\mathbf{1}^T \Sigma^{-1}(\lambda \mathbf{1} + \gamma \boldsymbol{\mu}) = 1 \tag{5}$$

$$\boldsymbol{\mu}^T \Sigma^{-1}(\lambda \mathbf{1} + \gamma \boldsymbol{\mu}) = r_{target} \tag{6}$$

Defining the following scalar quantities:

$$A = \mathbf{1}^T \Sigma^{-1} \mathbf{1} \tag{7}$$

$$B = \mathbf{1}^T \Sigma^{-1} \boldsymbol{\mu} = \boldsymbol{\mu}^T \Sigma^{-1} \mathbf{1} \tag{8}$$

$$C = \boldsymbol{\mu}^T \Sigma^{-1} \boldsymbol{\mu} \tag{9}$$

Equations (5) and (6) can be rewritten as the linear system:

$$\begin{bmatrix} A & B \\ B & C \end{bmatrix} \begin{bmatrix} \lambda \\ \gamma \end{bmatrix} = \begin{bmatrix} 1 \\ r_{target} \end{bmatrix} \tag{10}$$

We solve this system using Cramer's rule. The determinant is:

$$\Delta = \begin{vmatrix} A & B \\ B & C \end{vmatrix} = AC - B^2 \tag{11}$$

The solutions are:

$$\lambda = \frac{1}{\Delta}\begin{vmatrix} 1 & B \\ r_{target} & C \end{vmatrix} = \frac{C - r_{target}B}{AC - B^2} \tag{12}$$

$$\gamma = \frac{1}{\Delta}\begin{vmatrix} A & 1 \\ B & r_{target} \end{vmatrix} = \frac{Ar_{target} - B}{AC - B^2} \tag{13}$$

Substituting equations (12) and (13) into equation (4), we obtain the optimal weight vector:

$$\mathbf{w} = \Sigma^{-1}\left(\frac{C - r_{target}B}{AC - B^2}\mathbf{1} + \frac{Ar_{target} - B}{AC - B^2}\boldsymbol{\mu}\right) \tag{14}$$

However, we need to add the non-negativity constraints. Indeed, the weight vector $\mathbf{w}$ is not guaranteed to be non-negative, and we require a long-only portfolio (i.e., $w_i \geq 0$ for all i). To solve this Quadratic Programming (QP) problem, we define the Lagrangian $\mathcal{L}$ with multipliers λ (budget), γ (return), and $\mathbf{z}$ (non-negativity):

$$\mathcal{L}(\mathbf{w}, \lambda, \gamma, \mathbf{z}) = \frac{1}{2}\mathbf{w}^T\Sigma\mathbf{w} - \lambda(\mathbf{w}^T\mathbf{1} - 1) - \gamma(\mathbf{w}^T\boldsymbol{\mu} - r_{target}) - \mathbf{z}^T\mathbf{w} \tag{15}$$

The Karush-Kuhn-Tucker (KKT) first-order necessary conditions for optimality are:

1. **Stationarity:** $\nabla_{\mathbf{w}}\mathcal{L} = \Sigma\mathbf{w} - \lambda\mathbf{1} - \gamma\boldsymbol{\mu} - \mathbf{z} = 0$
2. **Primal Feasibility:** $\mathbf{w}^T\mathbf{1} = 1$ and $\mathbf{w}^T\boldsymbol{\mu} = r_{target}$
3. **Dual Feasibility:** $z_i \geq 0$
4. **Complementary Slackness:** $z_i w_i = 0, \quad \forall i$

3 Spectral Instability and Ledoit-Wolf Shrinkage

The sample covariance matrix S is often ill-conditioned. In the spectral domain, we represent S via its eigen-decomposition $S = V\Lambda V^T$. The weight vector is sensitive to the inverse of the eigenvalues λ_i. If $\lambda_{min} \to 0$, the weights $\mathbf{w}$ exhibit extreme variance and instability.

We implement the **Ledoit-Wolf Shrinkage** to regularize the spectrum. The estimator $\hat{\Sigma}_{LW}$ is a convex combination of S and a highly structured target $F = \bar{\sigma}^2 I$:

$$\hat{\Sigma}_{LW} = (1 - \delta)S + \delta F \tag{16}$$

By shifting the eigenvalues $\lambda_i^{new} = (1-\delta)\lambda_i^{raw} + \delta\bar{\sigma}^2$, we bound the condition number $\kappa(\hat{\Sigma}_{LW})$ and stabilize the KKT system, preventing "corner solutions" driven by statistical noise.

4 Derivation of the Optimal Shrinkage Intensity

4.1 Loss Function and First-Order Condition

The loss function $L(\delta)$ is defined as:

$$L(\delta) = E[\|\hat{\Sigma}(\delta) - \Sigma\|_{Fro}^2] = E[\mathrm{Tr}((\hat{\Sigma}(\delta) - \Sigma)^T(\hat{\Sigma}(\delta) - \Sigma))] \tag{17}$$

Defining the error terms $A = S - \Sigma$ and $B = F - S$, the estimator error can be rewritten as:

$$\hat{\Sigma}(\delta) - \Sigma = (S - \Sigma) + \delta(F - S) = A + \delta B \tag{18}$$

Expanding the trace via linearity:

$$L(\delta) = E[\mathrm{Tr}(A^T A)] + 2\delta E[\mathrm{Tr}(B^T A)] + \delta^2 E[\mathrm{Tr}(B^T B)] \tag{19}$$

The first-order condition $\frac{dL}{d\delta} = 0$ yields the theoretical optimal intensity:

$$\delta^* = -\frac{E[\mathrm{Tr}(B^T A)]}{E[\mathrm{Tr}(B^T B)]} \tag{20}$$

4.2 Decomposition into π, ρ, and γ Parameters

To make this solution computable, we decompose the numerator and denominator into three fundamental parameters.

The parameter γ (Distance to structure): This represents the expected energy of the displacement toward the target:

$$\gamma = E[\|F - S\|_{Fro}^2] = E[\mathrm{Tr}(B^T B)] \tag{21}$$

The parameter π (Sample noise): This corresponds to the sum of the variances of the elements in the sample matrix:

$$\pi = \sum_{i,j} \mathrm{Var}(s_{ij}) = E[\mathrm{Tr}(S^T S)] - \mathrm{Tr}(\Sigma^T \Sigma) \tag{22}$$

The parameter ρ (Stochasticity correction): Since F is constructed from S, it shares a portion of the sampling error. This term captures the covariance between the errors of the target and the sample:

$$\rho = \sum_{i,j} \mathrm{Cov}(f_{ij}, s_{ij}) = E[\mathrm{Tr}(F^T S)] - \mathrm{Tr}(F^T \Sigma) \tag{23}$$

4.3 Final Synthesis

Substituting these definitions into the expression for δ^*, we find:

$$E[\mathrm{Tr}(B^T A)] = E[\mathrm{Tr}((F - S)^T(S - \Sigma))] = \rho - \pi \tag{24}$$

Which leads to the optimal Ledoit-Wolf intensity:

$$\delta^* = \frac{\pi - \rho}{\gamma} \tag{25}$$

5 Implementation Interpretation

The intensity δ^* is the ratio of the net estimation error $(\pi - \rho)$ to the distance from the target structure γ. In our empirical application to French equities, if the market is highly volatile (high π) or the time series is short, the shrinkage becomes more aggressive ($\delta \to 1$).

The inclusion of ρ is vital: it ensures that if the target F drifts in the same direction as the error in S, the shrinkage intensity is reduced to avoid reinforcing a pre-existing statistical bias.

6 The Finite-Sample Estimator

In the population limit, the parameters π, ρ, and γ represent asymptotic quantities. However, for a practical implementation with a sample of size T, we must acknowledge that the estimation variance of the sample covariance entries s_{ij} is $O(1/T)$.

To maintain consistency with the Ledoit-Wolf (2004) framework, we define the practical shrinkage intensity as:

$$\hat{\delta}^* = \max\left(0, \min\left(1, \frac{1}{T}\frac{\hat{\pi} - \hat{\rho}}{\hat{\gamma}}\right)\right) \tag{26}$$

where $\hat{\pi}$, $\hat{\rho}$, and $\hat{\gamma}$ are consistent estimators of their respective population parameters.

This formulation yields a critical asymptotic result:

$$\lim_{T\to\infty} \hat{\delta}^* = 0 \tag{27}$$

This implies that as the number of observations T grows, the sample covariance S converges in probability to the true covariance Σ, and the necessity for regularization through the target F vanishes. This ensures that our optimizer remains statistically consistent while providing maximum stability in the high-dimensional regime where $N \approx T$.